Constant Spectral Data Do Not Control Matrix Schrödinger Maximal Regularity

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Status. Candidate full counterexample, likely valid, to pointwise-spectral reverse-Hölder criteria for the system maximal estimate. The packet also gives a sign-free positive repair under comparable eigenvalues. The source's broader search for a geometry-sensitive matrix condition remains open.

1 Source question

Bailey studies the class $\mathcal{W}_2$ associated with $H = -\Delta I_m + |V|$ and asks for a reverse-Hölder-type condition implying boundedness of $|V|(|V| - \Delta)^{-1}$ when $m > 1$ [1, Section 5.3, printed p. 42]. The exact source passage is shown below.

> In analogy to the scalar case, it is quite likely that a similar reverse Hölder type condition will be sufficient to imply the boundedness of the operator $|V|\left(|V| - \Delta\right)^{-1}$ for $m > 1$. However, as far as the author is aware, this is still an open problem for $m > 1$. What is apparent is that the condition that the potential belongs to $L^{\frac{n}{2}}$ will once again be sufficient to imply that it belongs to $\mathcal{W}_2$. The following proposition has an identical proof to that of Proposition 5.3.
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> **Proposition 5.4.** *For $n > 4$ and $m \in \mathbb{N}^*$,*
> $$L^{\frac{n}{2}}\left(\mathbb{R}^n; \mathcal{L}\left(\mathbb{C}^m\right)\right) \subset \mathcal{W}_2(\mathbb{R}^n; \mathcal{L}(\mathbb{C}^m)).$$

For a measurable Hermitian positive-semidefinite matrix potential $W : \mathbb{R}^n \to \mathcal{L}(\mathbb{C}^m)$, put

$$H = -\Delta I_m + W.$$

The assertion $W \in \mathcal{W}_2$ means

$$\|Wu\|_2 + \|\Delta u\|_2 \le C\|Hu\|_2, \qquad u \in D(H). \tag{1}$$

A first interpretation of the reverse-Hölder-type condition is to apply the scalar condition to $\|W(x)\|_{\mathrm{op}}$, a Schatten norm, the trace, or the ordered eigenvalue functions. The counterexample below rules out every condition depending only on this pointwise spectral data.

2 Proof intuition

The scalar obstruction is the unit-ball potential $v = \mathbf{1}_{B(0,1)}$ in $\mathbb{R}^3$. It has a bounded positive zero-energy solution for $-\Delta + v$. Cutting this solution off at radius R leaves a fixed potential term but creates only an $O(R^{-1/2})$ residual.

To make the matrix example spectrally featureless, place v in one diagonal channel and its complement $1-v$ in the other. The resulting matrix is a rank-one orthogonal projection at every point. Its eigenvalues, trace, operator norm, and all Schatten norms are constant, but the bad scalar channel survives unchanged. Thus the obstruction is the spatial motion of the eigenspaces, information that pointwise eigenvalues cannot see.

3 Full constant-spectrum counterexample

Theorem 1. *There is a bounded measurable real-symmetric potential $W : \mathbb{R}^3 \to \mathcal{L}(\mathbb{R}^2)$ such that, almost everywhere,*

$$W(x)^2 = W(x), \qquad \operatorname{rank} W(x) = 1, \qquad \operatorname{spec} W(x) = \{0,1\},$$

but $W \notin \mathcal{W}_2$.

Consequently, both ordered eigenvalues are constant, every Schatten norm of $W(x)$ is identically one, and

$$\operatorname{tr}(W(x)^k) = 1 \qquad (k = 1, 2, \ldots).$$

In particular, no hypothesis depending only on pointwise spectral data and satisfied by the constant spectrum $\{0,1\}$ can imply the matrix $\mathcal{W}_2$ estimate.

Proof. Let $B = B(0,1)$, $v = \mathbf{1}_B$, and set

$$W(x) = \begin{pmatrix} 1 - v(x) & 0 \\ 0 & v(x) \end{pmatrix}. \tag{2}$$

Since $v^2 = v$, the matrix in (2) is a rank-one orthogonal projection at every point away from the irrelevant boundary sphere. All asserted spectral identities follow immediately.

It remains to disprove (1). Put $a = 1 - \tanh 1$ and define

$$h(r) = \begin{cases} \dfrac{\sinh r}{r \cosh 1}, & 0 < r \le 1, \\ 1 - \dfrac{a}{r}, & r \ge 1, \end{cases} \qquad h(0) = \frac{1}{\cosh 1}.$$

At $r = 1$, both formulas have value $\tanh 1$ and derivative $a = (\cosh 1 - \sinh 1)/\cosh 1$. Hence no surface delta term occurs, and direct radial differentiation gives

$$(-\Delta + v)h = 0 \tag{3}$$

distributionally on $\mathbb{R}^3$.

Choose a fixed smooth radial cutoff η equal to one on $[0,1]$ and zero on $[2,\infty)$, and put

$$\eta_R(x) = \eta(|x|/R), \qquad u_R = \eta_R h, \qquad U_R = (0, u_R), \qquad R > 2.$$

The matching just checked implies $u_R \in H^2(\mathbb{R}^3)$, hence $U_R \in D(H)$. Since the second channel of W is v,

$$\|WU_R\|_2 = \|vu_R\|_2 = \|vh\|_2 > 0 \tag{4}$$

is independent of R. On the annulus $R < |x| < 2R$, equation (3) yields

$$HU_R = (0, -2\nabla\eta_R \cdot \nabla h - h\Delta\eta_R)\,.$$

There $h = O(1)$, $|\nabla h| = O(R^{-2})$, $|\nabla \eta_R| = O(R^{-1})$, and $|\Delta \eta_R| = O(R^{-2})$. The residual is therefore $O(R^{-2})$ on a set of volume $O(R^3)$, so

$$\|HU_R\|_2 = O(R^{-1/2}) \longrightarrow 0. \tag{5}$$

Equations (4)–(5) contradict $\|WU\|_2 \leq C\|HU\|_2$, proving $W \notin \mathcal{W}_2$. □

Corollary 1. *Let $\mathfrak{P}$ be any pointwise-unitarily-invariant condition on positive-semidefinite matrix fields that depends only on their ordered eigenvalue functions. If $\mathfrak{P}$ holds for the constant eigenvalue list $(0, 1)$, then $\mathfrak{P}$ does not imply membership in $\mathcal{W}_2$.*

Proof. The potential (2) has that eigenvalue list everywhere and fails $\mathcal{W}_2$ by Theorem 1. □

This includes the Bochner interpretation

$$\left(\frac{1}{|Q|}\int_Q \|W(x)\|^q \, dx\right)^{1/q} \leq \frac{C}{|Q|}\int_Q \|W(x)\| \, dx,$$

for any unitarily invariant matrix norm: both sides are the same constant.

4 A sign-free positive repair

The counterexample shows that some control of eigenspace geometry or of all quadratic-form directions is essential. One clean spectral repair is to prevent the eigenvalues from separating.

Theorem 2. *Let $W : \mathbb{R}^n \to \mathcal{L}(\mathbb{C}^m)$ be measurable and positive semidefinite. Write*

$$\lambda(x) = \lambda_{\min}(W(x)), \qquad \Lambda(x) = \lambda_{\max}(W(x)).$$

If, for some $q \geq 2$ and $\kappa \geq 1$,

$$\lambda \in RH_q, \qquad \Lambda \leq \kappa\lambda \quad a.e.,$$

then $W \in \mathcal{W}_2$. The constant depends only on the scalar RH_q data, n, and κ. No off-diagonal sign condition is required, and complex Hermitian potentials are allowed.

Proof. Let $h = -\Delta + \lambda$. Bailey's scalar theorem [1, Theorem 5.1], quoting Auscher–Ben Ali, gives uniformly for $\varepsilon > 0$,

$$\|\lambda(h+\varepsilon)^{-1}g\|_2 \leq C_s\|g\|_2. \tag{6}$$

For $F \in L^2(\mathbb{R}^n; \mathbb{C}^m)$, the vector Kato inequality and $W \geq \lambda I_m$ give the pointwise resolvent domination

$$|(H+\varepsilon)^{-1}F| \leq (h+\varepsilon)^{-1}|F|. \tag{7}$$

This uses no cooperativity or entrywise sign assumption.

Since $|Wz| \leq \Lambda|z| \leq \kappa\lambda|z|$, equations (6)–(7) imply

$$\|W(H+\varepsilon)^{-1}F\|_2 \leq \kappa C_s\|F\|_2.$$

For $u \in D(H)$, take $F = (H+\varepsilon)u$, then let $\varepsilon \downarrow 0$. This yields $\|Wu\|_2 \leq \kappa C_s\|Hu\|_2$. Finally, $-\Delta u = Hu - Wu$, which proves (1). □

5 The geometry-sensitive frontier

Davey and Isralowitz define a matrix reverse-Hölder class $\mathcal{B}_q$ by requiring

$$\left(\frac{1}{|Q|}\int_Q \langle W(x)e,e\rangle^q\,dx\right)^{1/q} \le C\left\langle\left(\frac{1}{|Q|}\int_Q W(x)\,dx\right)e,e\right\rangle$$

uniformly over cubes Q and constant vectors e [2, Definition 2.1]. This condition detects eigenspace motion and is not contradicted by Theorem 1: for $e=(0,1)$ the quadratic form is $\mathbf{1}_B$, and for $e=(1,0)$ it is $\mathbf{1}_{\mathbb{R}^3\setminus B}$; neither belongs to scalar RH_q.

Addona, Leone, Lorenzi and Rhandi prove broader L^p maximal estimates under comparable eigenvalues, a reverse-Hölder smallest eigenvalue, and nonpositive off-diagonal entries [3, Theorem 6.1]. Theorem 2 removes the sign assumption at $p=2$.

Thus the exact remaining frontier is geometry-sensitive: for example, whether membership in the directional matrix class $\mathcal{B}_2$ alone implies $\mathcal{W}_2$ is not settled by this packet.

6 Verification, search bounds, and recommendation

The script `code/verify_cutoff_counterexample.py` checks the projection identities, the matching conditions at $r=1$, and the exact radial cutoff residual. For radii from 8 to 8192, $\sqrt{R}\,\|HU_R\|_2$ stabilizes near 23.206, while $\|WU_R\|_2 = 1.4650804067$ remains fixed. The proof itself is analytic.

A bounded search through 27 August 2026 covered the run indexes, the exact source question, arXiv and web queries for matrix-valued Schrödinger maximal regularity, projection-valued potentials, constant spectrum, and matrix reverse-Hölder classes, plus source-level inspection of arXiv:2207.05790 and arXiv:2401.00479. No occurrence of the displayed constant-spectrum counterexample or an equivalent $\mathcal{W}_2$ obstruction was found. Novelty confidence is moderate because the construction is elementary.

Human review should check the interpretation boundary in Corollary 1; the analytic failure of $\mathcal{W}_2$ is explicit. The packet should be treated as a full negative result for pointwise-spectral criteria, not as a characterization of all matrix reverse-Hölder hypotheses.

References

[1] J. Bailey, *The Kato Square Root Problem for Divergence Form Operators with Potential*, J. Fourier Anal. Appl. 26 (2020), article 46; arXiv:1812.10196.

[2] B. Davey and J. Isralowitz, *Exponential Decay Estimates for Fundamental Matrices of Generalized Schrödinger Systems*, arXiv:2207.05790 (2022), Definition 2.1.

[3] D. Addona, V. Leone, L. Lorenzi, and A. Rhandi, *L^p Maximal Regularity for Vector-Valued Schrödinger Operators*, arXiv:2401.00479 (2023).